

Salil Patki

Naperville, IL · College Station, TX

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Education

Texas A&M University, College Station, TX

Expected May 2030

B.S. in Mechanical Engineering

- Member of Craig and Galen Bowen Engineering Honors
- President's Endowed Scholar Recipient (~250 of 15,000 incoming freshmen)
- National Merit Scholar (top ~0.5% nationally)
- Member of American Society of Mechanical Engineers (ASME), Solar Car Racing, Robomaster Robotics

Technical Skills

- **Design & CAD:** CAD/CAM (OnShape, Fusion 360, SolidWorks, FreeCAD), Design for Manufacturing (DFM), Mechanical Assembly Design, Tolerancing
- **Simulation & Analysis:** Computational Fluid Dynamics (CFD), OpenFOAM (OpenCFD), Underwater/Submersible Design, Failure Analysis & Troubleshooting, Finite Element Analysis (FEA), Engineering Physics
- **Manufacturing & Prototyping:** 3D Printing (FDM, SLA), CNC Machining, Waterjet Cutting
- **Robotics:** ROV & Competition Robot Design, Custom Drivetrains, Actuation & Mechanism Design, Machine Vision, Iterative Prototyping, Raspberry Pi, Arduino, Engineering Report Writing, Data Acquisition & Analysis
- **Programming:** Java, C++, Python
- **Tools:** Microsoft Office Suite, LaTeX

Robotics & Engineering Projects

FIRST Tech Challenge – Robotics Design & Team Lead

CAD · CNC · FDM · Waterjet

- Designed a **custom omnidirectional drivetrain** for full holonomic movement, enabling precise multi-axis maneuvering
- Engineered an **indexer** to reliably feed and sequence elements, and a **launcher** for consistent launch velocity and accuracy
- Integrated CNC-machined, waterjet-cut, and 3D-printed components into functional, serviceable subsystems
- Iterated through multiple prototypes to improve reliability and performance across competition cycles
- Led a multidisciplinary team through design reviews, build cycles, and competition testing

SeaPerch Underwater ROV Competition

CAD · FDM · SLA · CFD (OpenFOAM) · Systems Design

- Ranked **4th out of 9,000+ teams worldwide** and won the **Innovation Award** for the best ROV design in competition
- Designed a custom **ballasting system** to achieve neutral buoyancy and stable depth control underwater
- Integrated a **vision system** for underwater navigation and task completion
- Engineered **venturi-effect jet propulsion** to improve thrust efficiency and maneuverability
- Applied computational fluid dynamics (CFD) in OpenFOAM to optimize hull hydrodynamics, drag reduction, and stability
- Fabricated a waterproofed, hydrodynamic ROV chassis using additive manufacturing and iterative testing

USV Speed Record Attempt

CAD · Composites · CFD · CNC · Electrical Design · Thermal Management

- Architected a high-speed unmanned surface vehicle from a **first-principles power analysis**, deriving thrust, shaft power, and current requirements from aerodynamic and hydrodynamic drag estimates
- Designed a **twin counter-rotating surface drivetrain** to halve per-channel current, distribute thermal load, and cancel reaction torque for improved directional stability
- Modeled the hull as a **three-point hydroplane** in CAD, treating center-of-gravity placement and active stabilization as primary design parameters to mitigate high-speed aerodynamic lift and blowover
- Applied **computational fluid dynamics (CFD)** to refine drag estimates and characterize aerodynamic lift
- Engineered a **dual-loop ram-fed water-cooling system** and a **carbon-fiber/foam-core composite hull** with Kevlar-reinforced hardpoints at the transom, strut, and motor mounts

Awards & Honors

- National Merit Scholar – National Merit Scholarship Program
- Presidential Scholarship – Texas A&M University
- Excellence in Youth Leadership – City of Aurora
- Commendation – Illinois General Assembly
- IP Scholar – Indian Prairie School District